# SAL INSTITUTE OF TECHNOLOGY AND ENGINEERING RESEARCH

Opp. Science city, Sola-Bhadaj Road, Ahmedabad, Gujarat 380060

Ph: 079 67129000 | Website: www.sal.edu.in


---

# Student's Weekly Report of Internship

| | |
|---|---|
| **Student Name:** | Jainil Gupta |
| **Enrollment Number:** | 220670132018 |
| **Week Duration:** | January 29, 2026 - February 4, 2026 (Week 5) |
| **Name Of Organization:** | Microsoft Elevate Program (via Edunet Foundation & FICE Education) |
| **External Guide Name:** | Adarsh Gupta |
| **External Guide Contact:** | 0124-4080107 |
| **Internal Faculty Guide:** | Chintan Rana |

## Work done in last Week with description:

### 1. Week 5 Module: Supervised Learning

#### Description:

Attended comprehensive session on "Supervised Learning" on February 20, 2026. Learned about supervised learning algorithms, their applications, and hands-on implementation using scikit-learn library.

#### Key Topics Covered:

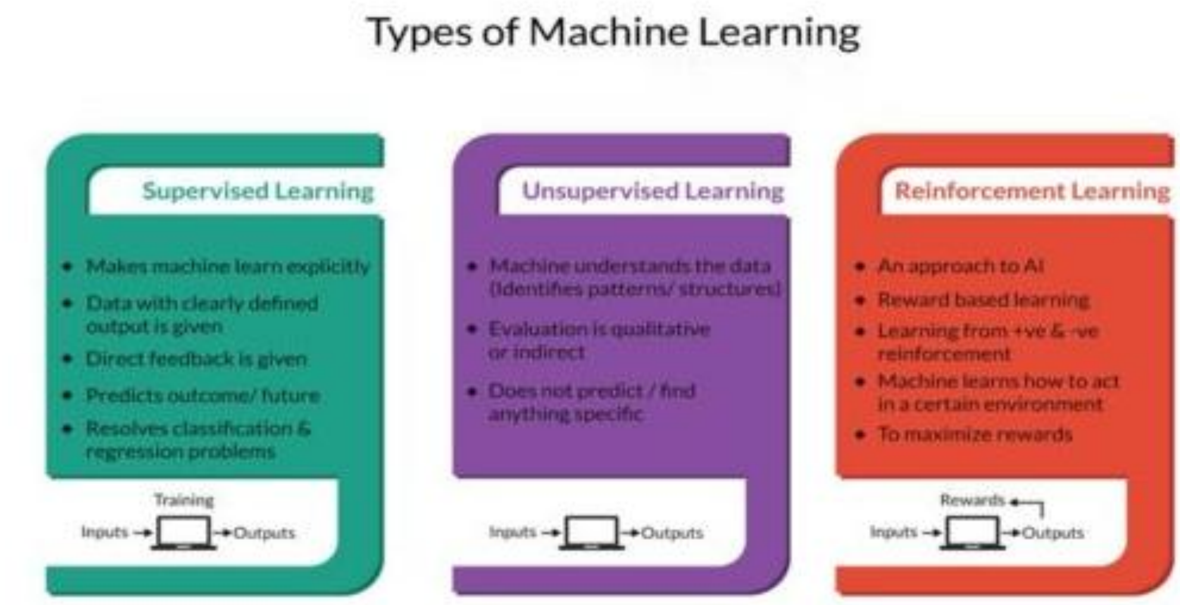
- Supervised learning concept: learning from labeled data
- Regression vs Classification problems
- Linear Regression: theory and implementation
- Logistic Regression for classification tasks
- Decision Trees: structure, splitting criteria, pruning
- Random Forest ensemble method
- Support Vector Machines (SVM) basics
- K-Nearest Neighbors (KNN) algorithm
- Train-test split and cross-validation techniques

#### Hands-on Practice:

Implemented Linear Regression on Boston Housing dataset. Trained a classification model using Iris dataset with Decision Tree and Random Forest algorithms. Learned to evaluate models using accuracy, precision, recall, and F1-score metrics. Practiced splitting data into training and testing sets (80-20 split).

#### Code Practice:

Used scikit-learn library to import datasets, preprocess data, train models, and evaluate performance. Practiced hyperparameter tuning and compared different algorithm performances.




## 2. Week 5 Module: Unsupervised Learning

### Description:

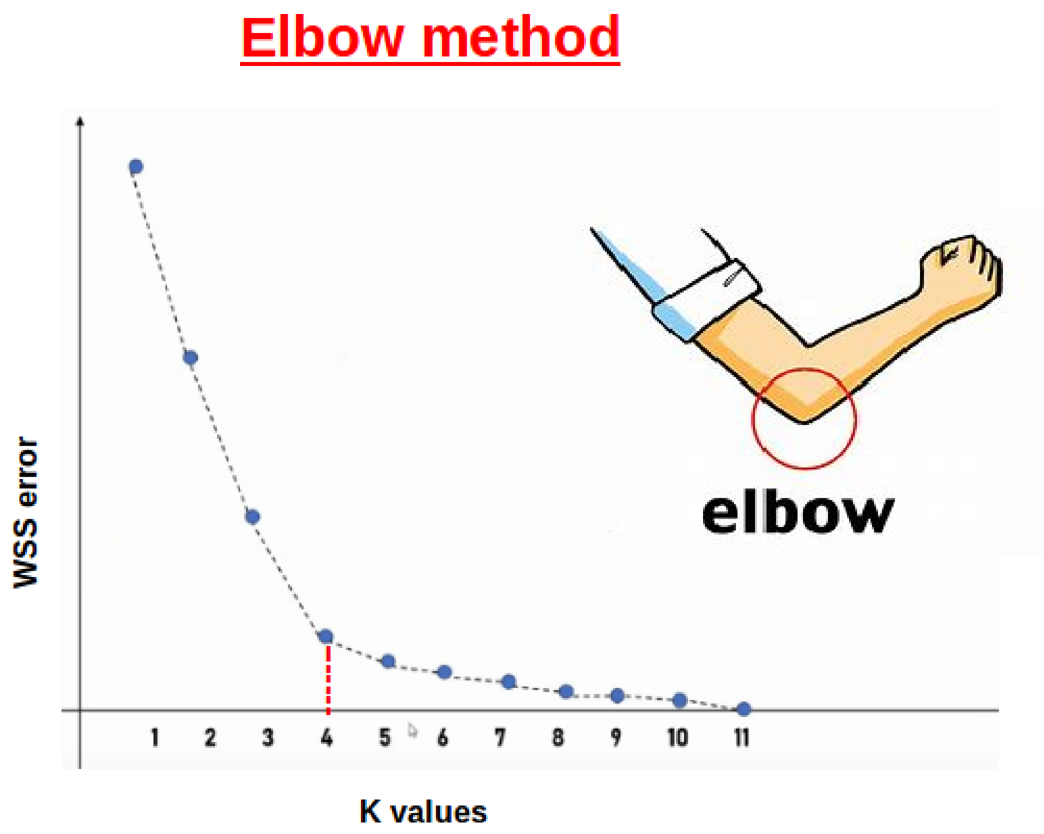
Participated in "Unsupervised Learning" session on February 23, 2026. Explored clustering algorithms, dimensionality reduction techniques, and their practical applications in data analysis.

### Key Topics Covered:

- Unsupervised learning: learning from unlabeled data
- Clustering algorithms and applications
- K-Means clustering: algorithm, initialization, convergence
- Hierarchical clustering: agglomerative vs divisive
- DBSCAN (Density-Based Spatial Clustering)
- Dimensionality reduction techniques
- Principal Component Analysis (PCA)
- Elbow method for optimal cluster selection
- Anomaly detection using clustering techniques

### Practical Implementation:

Applied K-Means clustering on customer segmentation dataset. Used Elbow method to determine optimal number of clusters. Implemented PCA for dimensionality reduction on high-dimensional data. Visualized clusters using matplotlib and seaborn libraries. Understood how clustering can be used for anomaly detection, which is relevant for surveillance project.




## 3. Week 5 Module: Reinforcement Learning

### Description:

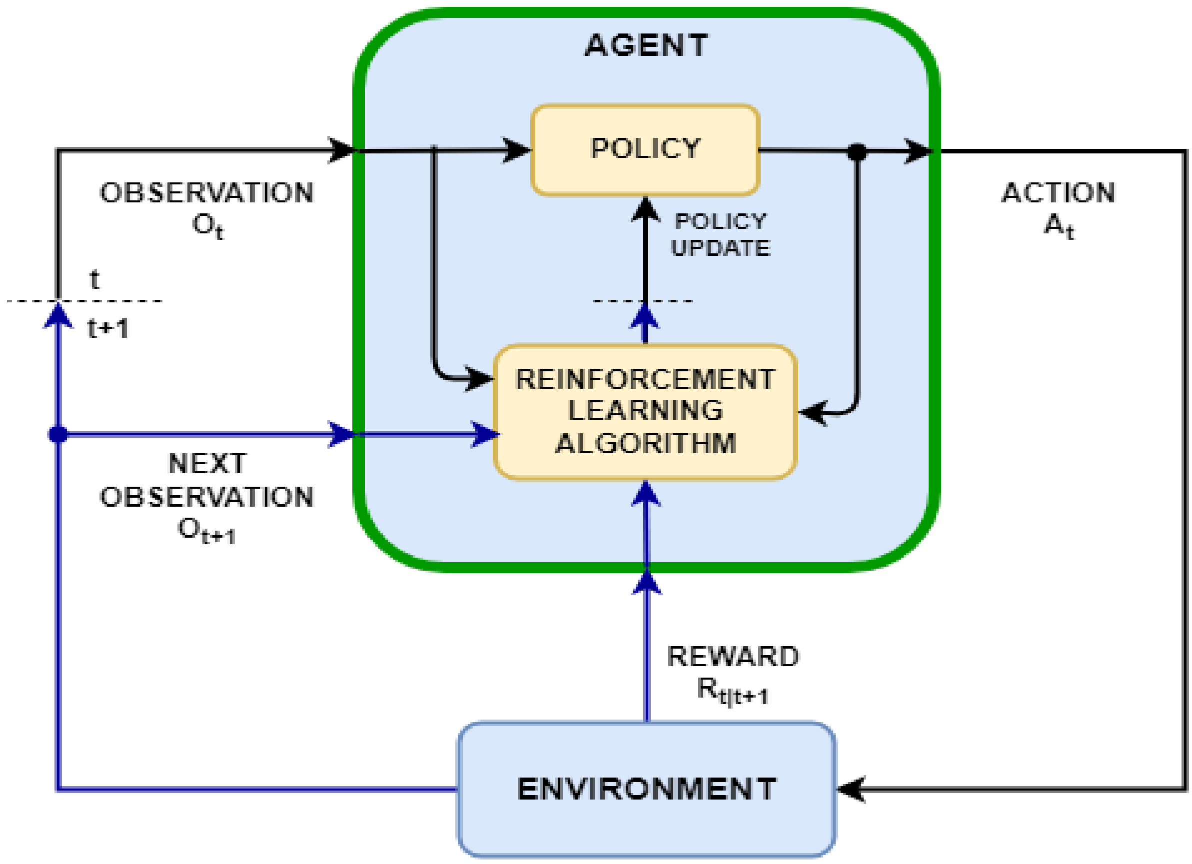
Attended "Reinforcement Learning" session on February 25, 2026. Learned about RL fundamentals, agent-environment interaction, and explored basic RL concepts with simple examples.

### Key Topics Covered:

- Reinforcement Learning fundamentals: agent, environment, actions, rewards
- Markov Decision Process (MDP) framework
- Exploration vs Exploitation trade-off
- Q-Learning algorithm basics
- Policy: deterministic vs stochastic
- Value functions and Bellman equation
- Applications: game playing, robotics, autonomous systems
- Difference between RL and Supervised/Unsupervised learning

### Examples and Demonstrations:

Explored classic RL examples like grid world navigation and game playing agents. Understood how RL agents learn optimal policies through trial and error. Discussed real-world RL applications in recommendation systems, robotics, and autonomous vehicles. Recognized that RL is more advanced and typically not used in basic object detection tasks.




## 4. Machine Learning Assignment Completion

### Description:

Completed comprehensive Machine Learning assignment covering supervised and unsupervised learning techniques. Implemented multiple algorithms and compared their performance on different datasets.

### Assignment Tasks:

- Task 1: Linear Regression on housing price prediction dataset
- Task 2: Classification using Decision Tree and Random Forest on Iris dataset
- Task 3: K-Means clustering on customer segmentation data
- Task 4: Model evaluation using confusion matrix, ROC curves
- Task 5: Hyperparameter tuning using GridSearchCV
- Created Jupyter notebook with complete code and visualizations
- Submitted assignment to MS Elevate portal

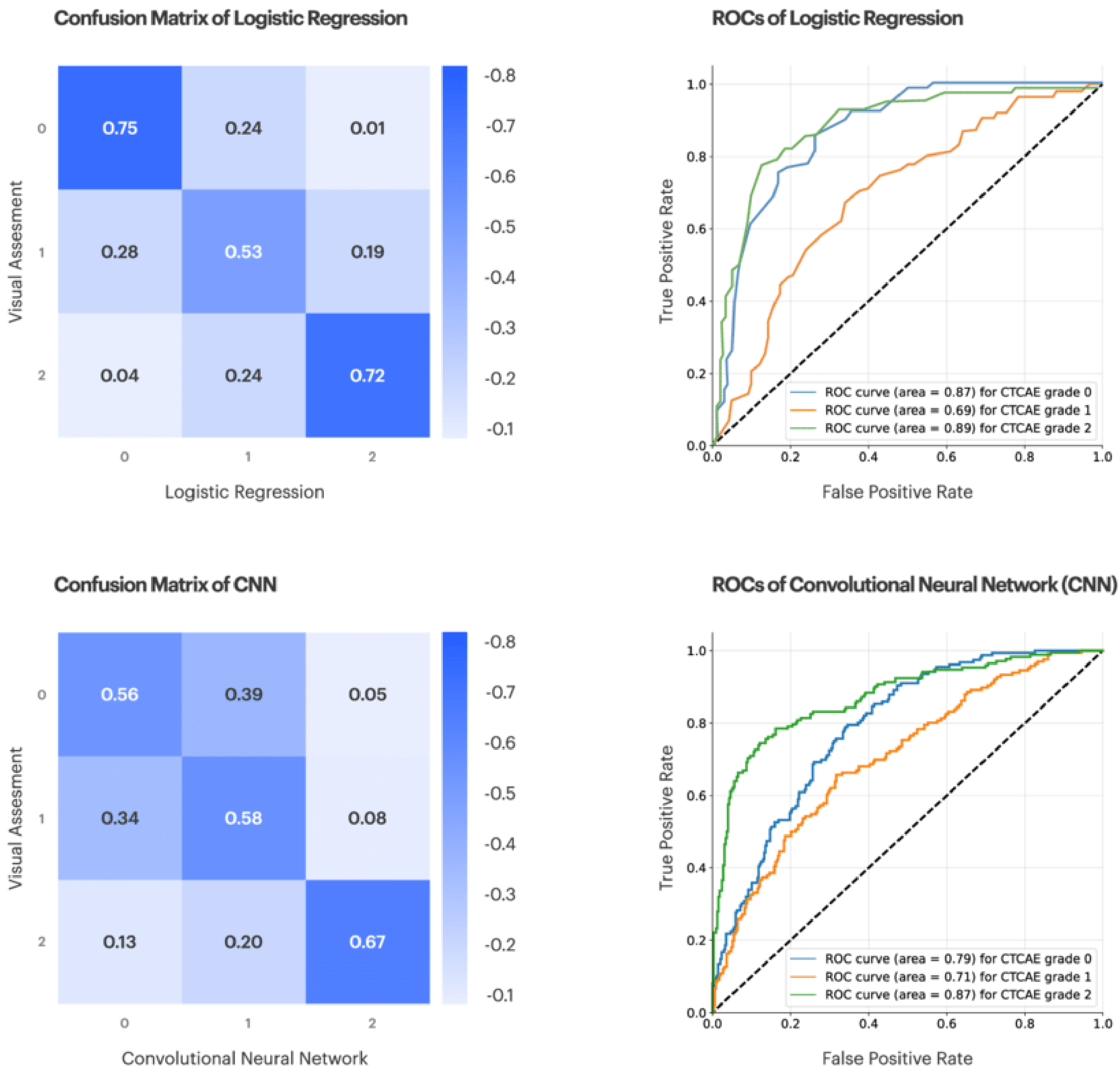



Figure: Example of confusion matrix and ROC curve used for evaluating classification model performance.

## 5. Scikit-learn Library Deep Dive

### Description:

Spent dedicated time exploring scikit-learn library documentation and practicing various ML workflows. Built strong foundation in using scikit-learn for future project implementation.

### Skills Developed:

- Data preprocessing: StandardScaler, MinMaxScaler, LabelEncoder
- Train-test split and cross-validation (KFold, StratifiedKFold)
- Model training: fit(), predict(), score() methods
- Model evaluation metrics: accuracy_score, precision_recall_fscore_support
- Pipelines for streamlined ML workflows
- Feature selection and extraction techniques

## 6. GTU Project - GitHub Repository Setup

### Description:

Created GitHub repository for the Border Surveillance project and setup initial project structure. Minimal development work as focus remained on completing ML modules this week.

### Setup Completed:

- Created repository: github.com/[username]/Border-Surveillance-AI
- Initialized with README.md, LICENSE (MIT), and .gitignore
- Created folder structure: data/, notebooks/, src/, models/, docs/
- Wrote project description in README with objectives and tech stack
- Added requirements.txt with essential libraries

### Note:

Deep Learning and CNN modules in Week 6-7 are essential prerequisites for implementing YOLOv8 object detection. Project development will accelerate after completing these modules.

## Plans for next week:

### 1. Complete Week 6 Microsoft Elevate Modules

- Attend "Deep Learning" fundamentals session (Feb 27)
- Participate in "Artificial Neural Networks (ANN)" session (Mar 2)
- Learn neural network architecture and backpropagation
- Practice building neural networks with TensorFlow/Keras

### 2. Install and Setup Deep Learning Environment

- Install TensorFlow 2.x and Keras
- Setup GPU support (if available) or use Google Colab
- Test basic neural network training
- Learn Keras Sequential and Functional API

### 3. Neural Network Fundamentals Practice

- Study activation functions: ReLU, Sigmoid, Tanh, Softmax
- Understand forward and backward propagation
- Practice with loss functions and optimizers
- Build simple feedforward neural network on MNIST dataset

### 4. Prepare for CNN Module (Week 7)

- Read about Convolutional Neural Networks basics
- Understand concepts: convolution, pooling, feature maps
- Study popular CNN architectures: LeNet, VGG, ResNet
- Prepare questions for CNN session

### 5. Update Project Documentation

- Add project architecture diagram to repository
- Document ML learning progress and key concepts
- Plan YOLOv8 implementation strategy for Week 7-8

**Total Working Hours:** 34 hours

Signature of Student: ______________________

**The above entries are correct, and the grading of work done by Trainee is:**

| Excellent | Very Good | Good | Fair | Below Average | Poor |
|---|---|---|---|---|---|
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

Signature of External Guide with Company Seal:

FICE EDUCATION
AUTHORISED SIGNATORY
ADARSH GUPTA
Microsoft Elevate

Date: 04/02/2026

Signature of Internal Guide: ______________________

Date: 04/02/2026